NTHU iGEM

AeroSense: Turning smell into data

THE PROBLEM

Spoilage smells before it shows

Diseased produce releases volatile organic compounds (VOCs) days to weeks before any visible symptom. Today's tools miss that window.

Visual inspection
Fast, but sees only symptoms. By then the batch has often shipped.

Lab assays (PCR, GC-MS)
Accurate, but slow, destructive and off-site.

TAIWAN BANANA SECTOR

NT$3.5B

estimated yearly fungal-disease loss risk, about a third of the sector's NT$10.4B output

OUR SOLUTION

Borrow the fruit fly's nose and brain

01 · WET LAB

Nose

Fly olfactory receptors in human cells turn odor into light

02 · MODEL

Brain

A fly mushroom-body circuit turns light into a decision

03 · HARDWARE

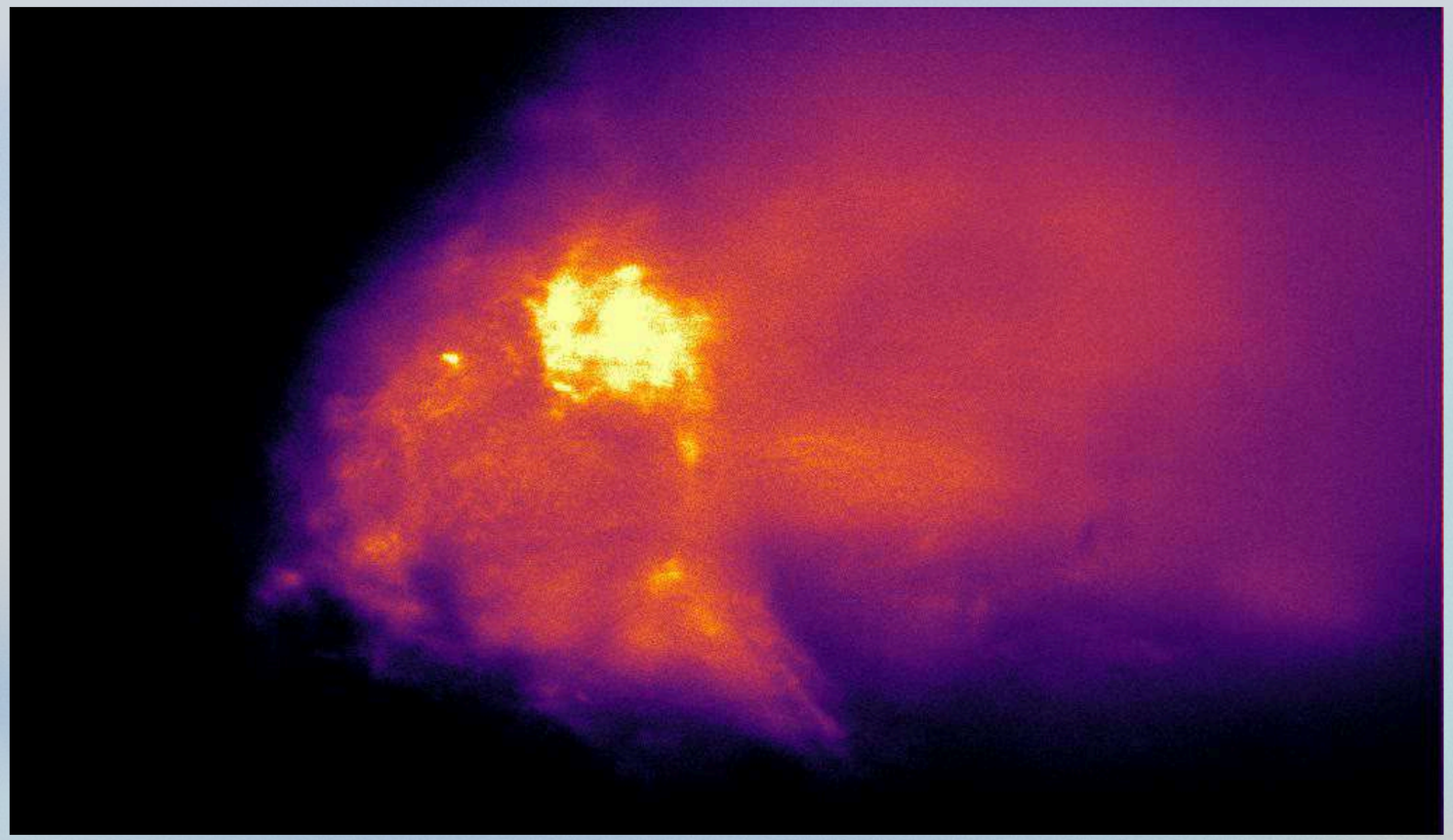
Device

Lock-in optics pull a faint glow out of real-world noise

Human Practices shaped every layer

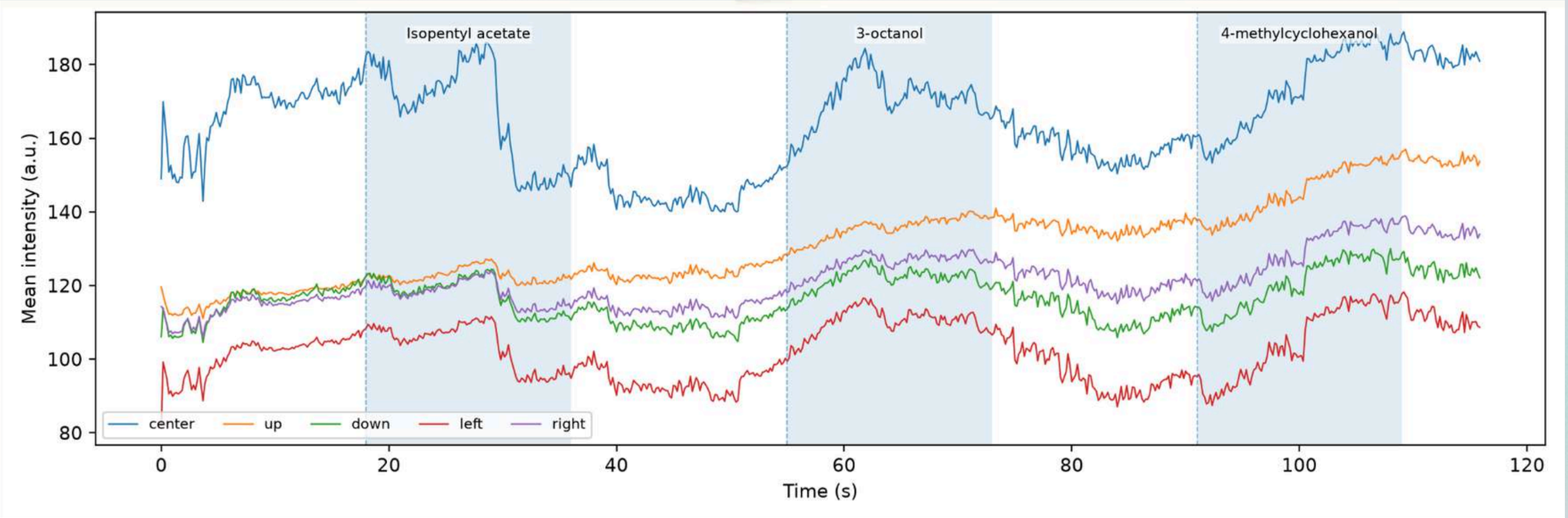
WET LAB · RECEPTOR LIBRARY

Before we start our project...



WET LAB · RECEPTOR LIBRARY

Before we start our project...



Readout: GCaMP fluorescence, 5 regions

Clearest rise: 3-octanol window

Next: $\Delta F/F_0$, solvent control, target VOCs

WET LAB

Can a living cell smell?



WET LAB · SENSOR DESIGN

A living cell that glows when it smells

1
VOC binds
to a fly olfactory receptor (OR)

→

2
Channel opens
OR + Orco form an ion channel

→

3
Ca^{2+} flows in
calcium enters the cell

→

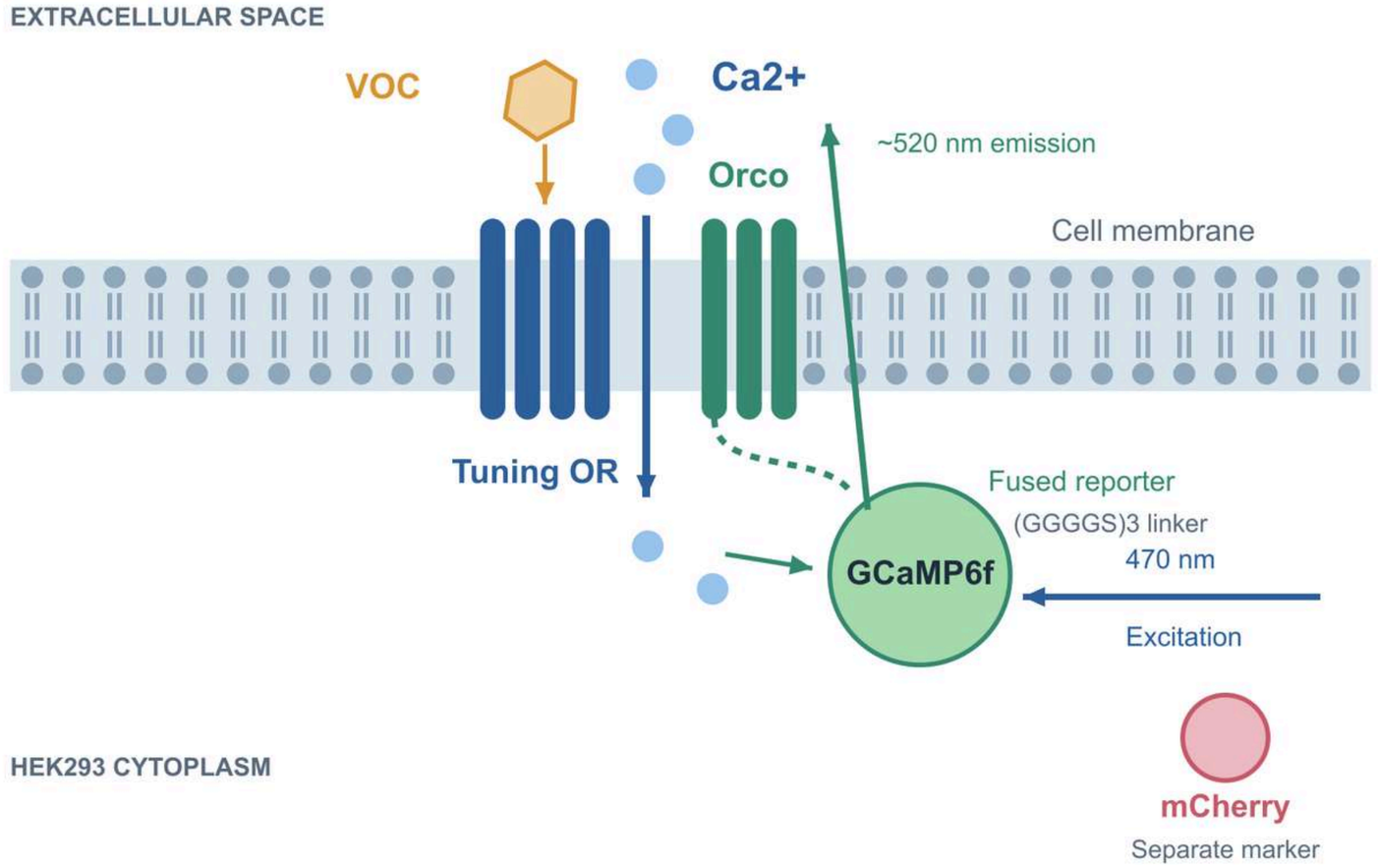
4
Cell glows
GCaMP6 fluoresces: stronger smell, brighter cell

Chassis: HEK293
A well-characterized human cell line. No live insects at any point.

Self-renewing
Cells keep making fresh receptors, so there is no chemical film to age or drift.

From odor recognition to green fluorescence

The biological design connects a fly receptor to an optical readout.



1 Recognize

A matching odorant binds OR.

2 Transduce

OR–Orco links binding to Ca2+ entry.

3 Report

GCaMP6f fluorescence tracks Ca2+.

4 Read

Blue excitation; green emission.

Mechanistic design; not a measured VOC response.

Design schematic • team Design page; mechanism: Zboray et al., 2023, doi:10.1038/s41598-023-47455-4

WET LAB / DESIGN DETAIL

Two modules, two different jobs

Red fluorescence marks expression; green fluorescence is the intended calcium readout.

MODULE 1

Receptor identity + expression marker



Two separate proteins: odorant receptor and red mCherry marker.

MODULE 2

Calcium reporter fused to the co-receptor



Shared pcDNA3.1(+) backbone. Schematic only; sequence lengths are not to scale.

WET LAB · RECEPTOR LIBRARY

Seven receptors, two early leads

Candidate ORs chosen from a ligand-profiling study of plant and plant-pathogen volatiles

Or10a · **Or13a** · Or47a · **Or49b** · Or71a · Or85b · Or98a

Zboray et al., 2023, Scientific Reports

BACTERIAL LEAD

Or49b → indole

Indole is a classic marker of E. coli contamination. Or49b is already cloned and sequenced.

FUNGAL LEAD

Or13a → 1-octen-3-ol

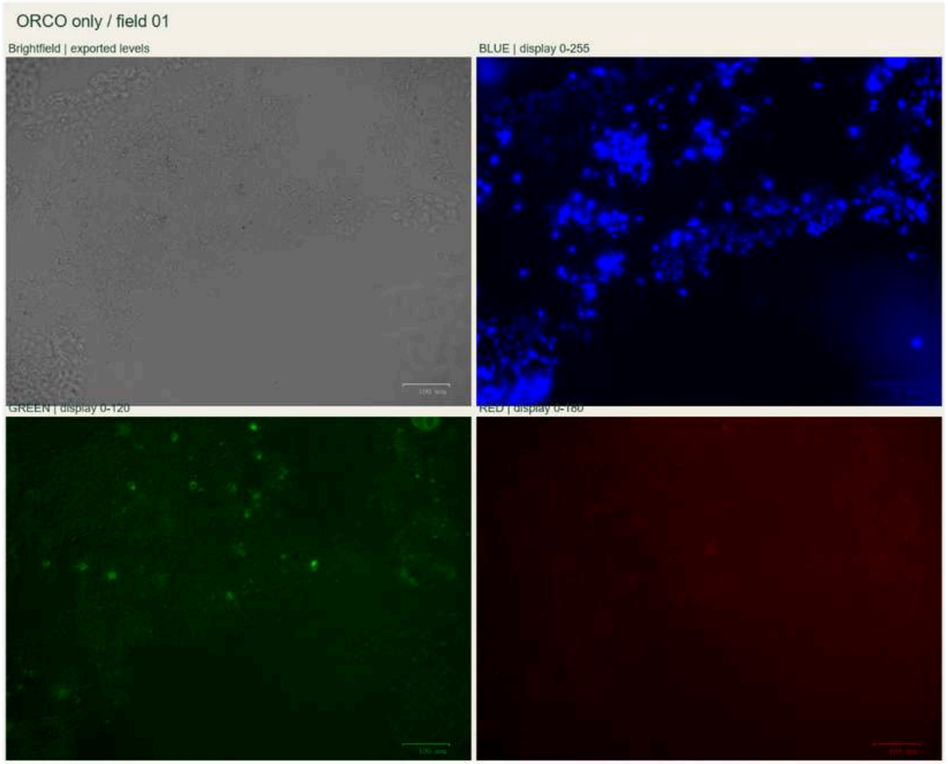
"Mushroom alcohol", a fungal volatile, fits our crop-disease focus.

WET LAB / OBSERVED EXPRESSION

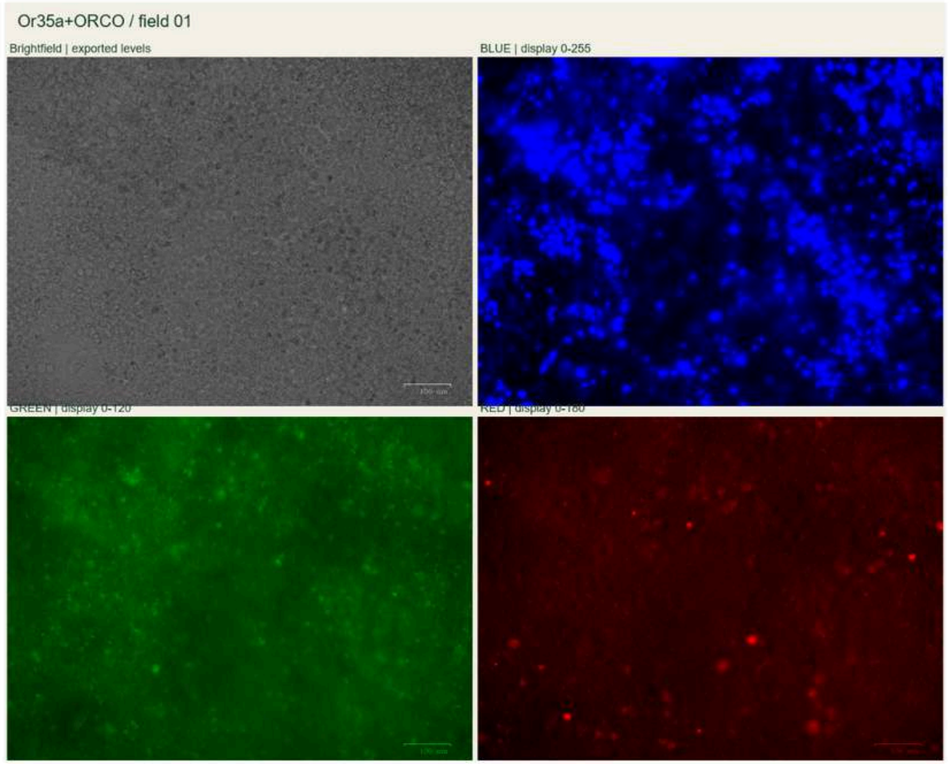
First, check whether the reporters are visible

Matched brightfield, DAPI, GREEN and RED channels; source display settings and scale bars retained.

ORCO only • field 01



Or35a + ORCO • field 01



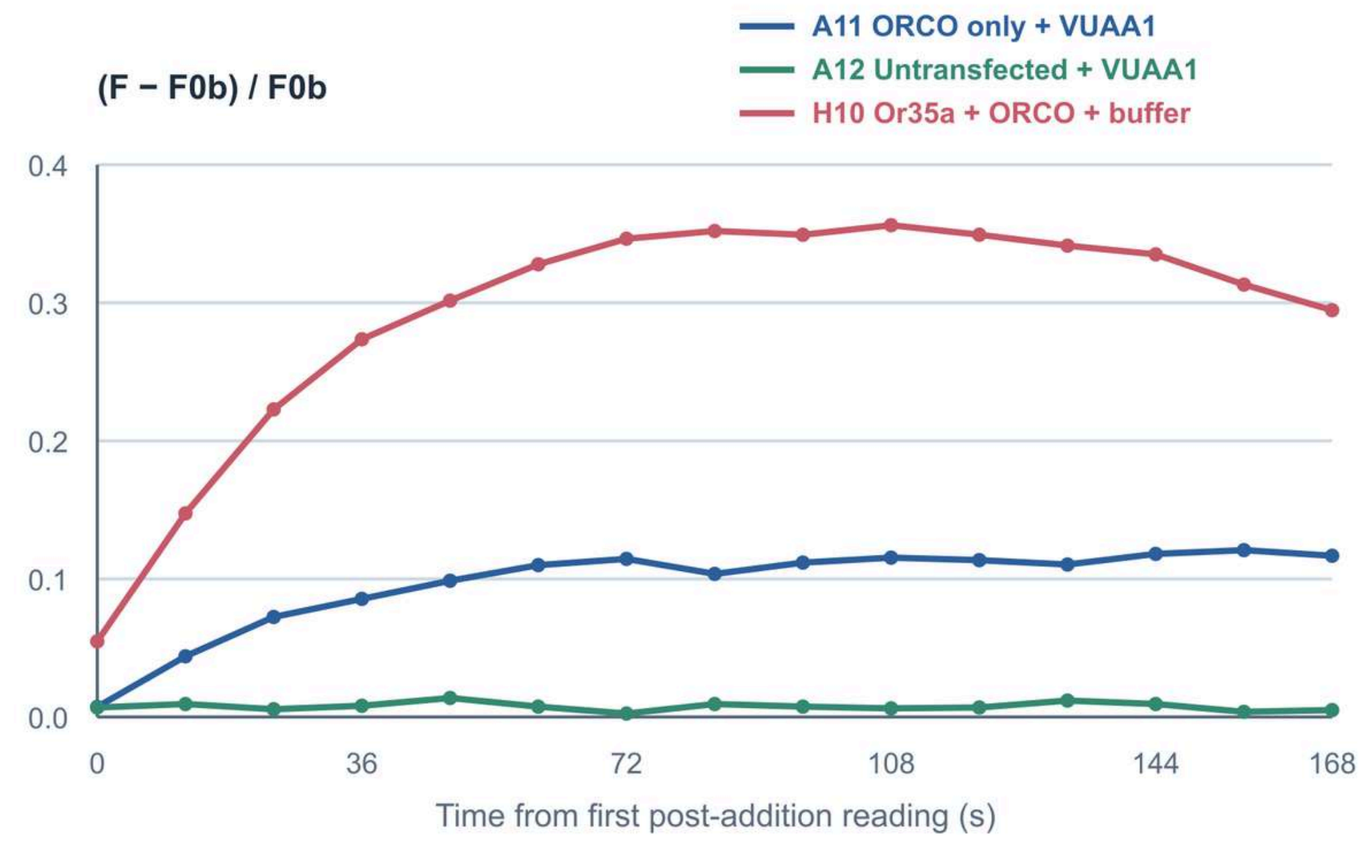
Single study • 44–48 h after transfection • before selection • illustrative fields, not biological replicates

Source: team microscopy WL-MIC-01 • complete field-01 panels; no intensity adjustment applied here

WET LAB / MEASURED PILOT

A fluorescence rise worth following up

Plate P2 • 10–11 October 2026 • approximately 72 h after transfection • before G418 selection



Excursion = maximum 3-reading mean / paired baseline − 1. Curves show all 15 test readings.

What we observed

+11.86%

A11 VUAA1 test excursion

+0.98%

A12 untransfected comparator

+35.25%

H10 buffer-only excursion

Promising observation,
still needs a matched-vehicle repeat.

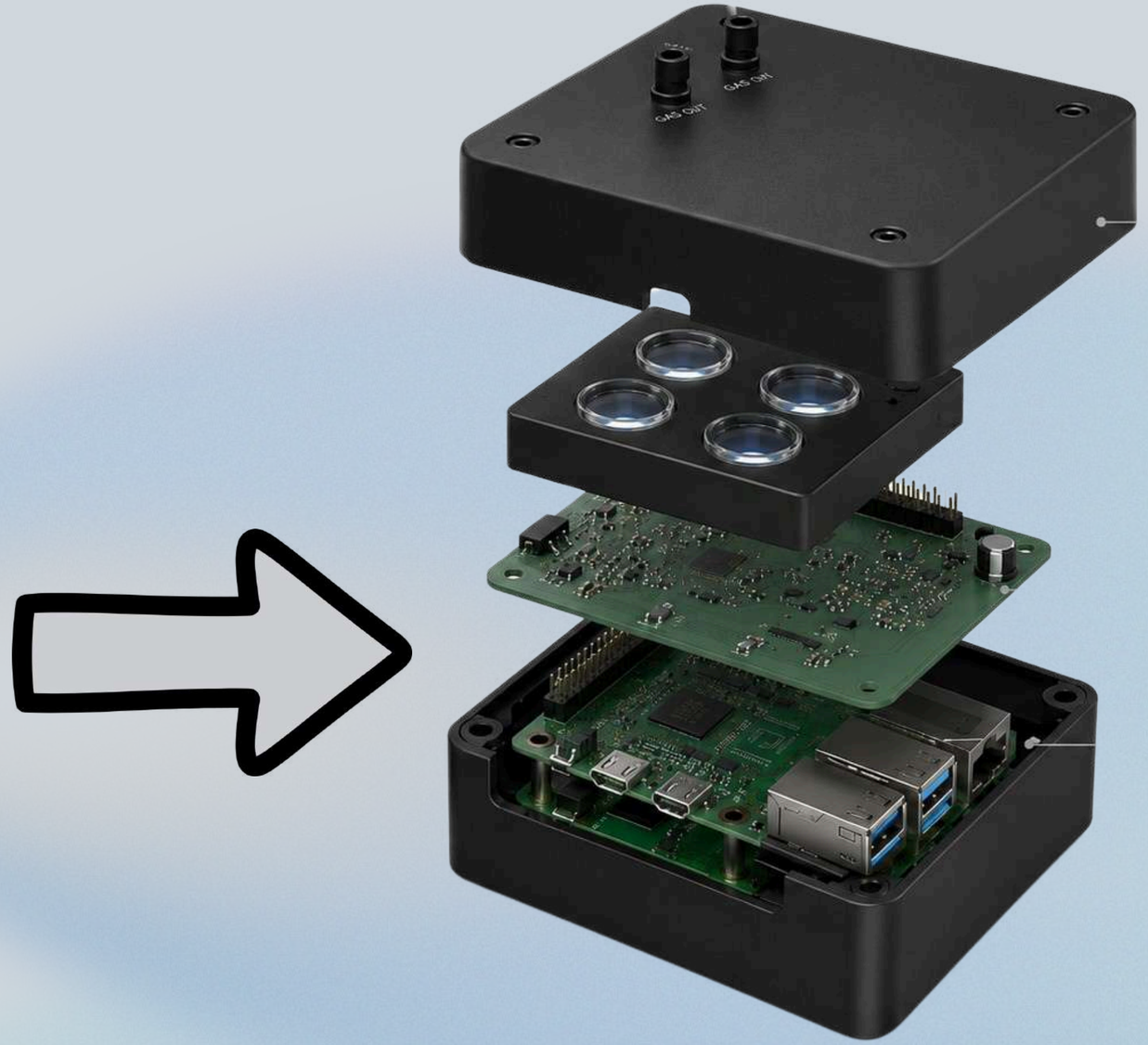
One well per condition × stimulus.
No matched DMSO-only control.
No receptor-specific conclusion.

Source: P2 raw-derived time series and well summary • VUAA1 nominal 50 μM • time zero is not the addition time

MODEL

The cell lit up. What did it smell?




GAS OUT
GAS OUT

MODEL · NEUROMORPHIC DECODER

Decoding with a fly's brain

PN

PROJECTION NEURONS

Lateral inhibition boosts signal-to-noise

→

KC

KENYON CELLS

Sparse coding separates overlapping odors

→

MBON

OUTPUT NEURONS

Make the call: which odor, and is it a risk?

Built on our advisors' own published work

Cheng, Charng, … Lo, Lee (2025). Hybrid neural networks in the mushroom body drive olfactory preference in Drosophila. Science Advances 11(22).

MODEL · WHY A FLY BRAIN

Small enough to run at the edge

<16 ms

decoding latency per reading

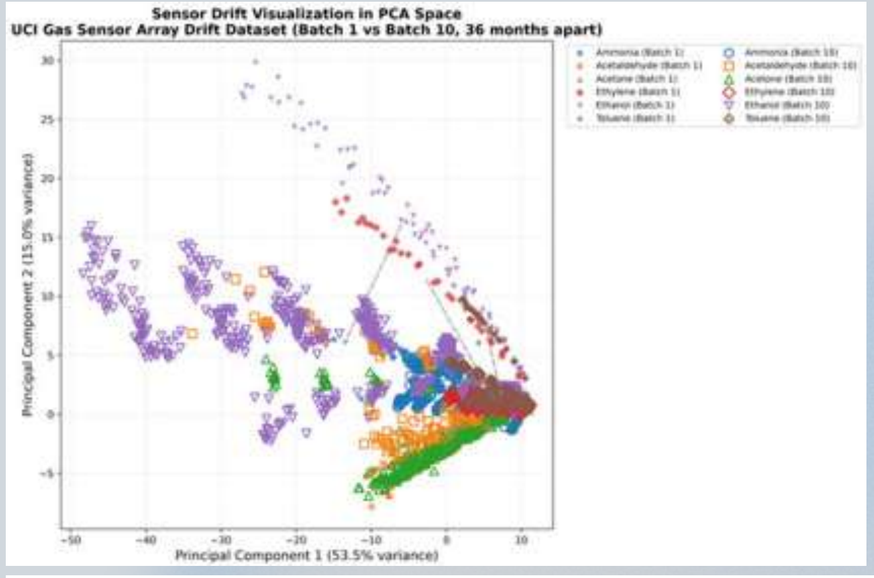
<1 W

sensing to classification on a Raspberry Pi

>85%

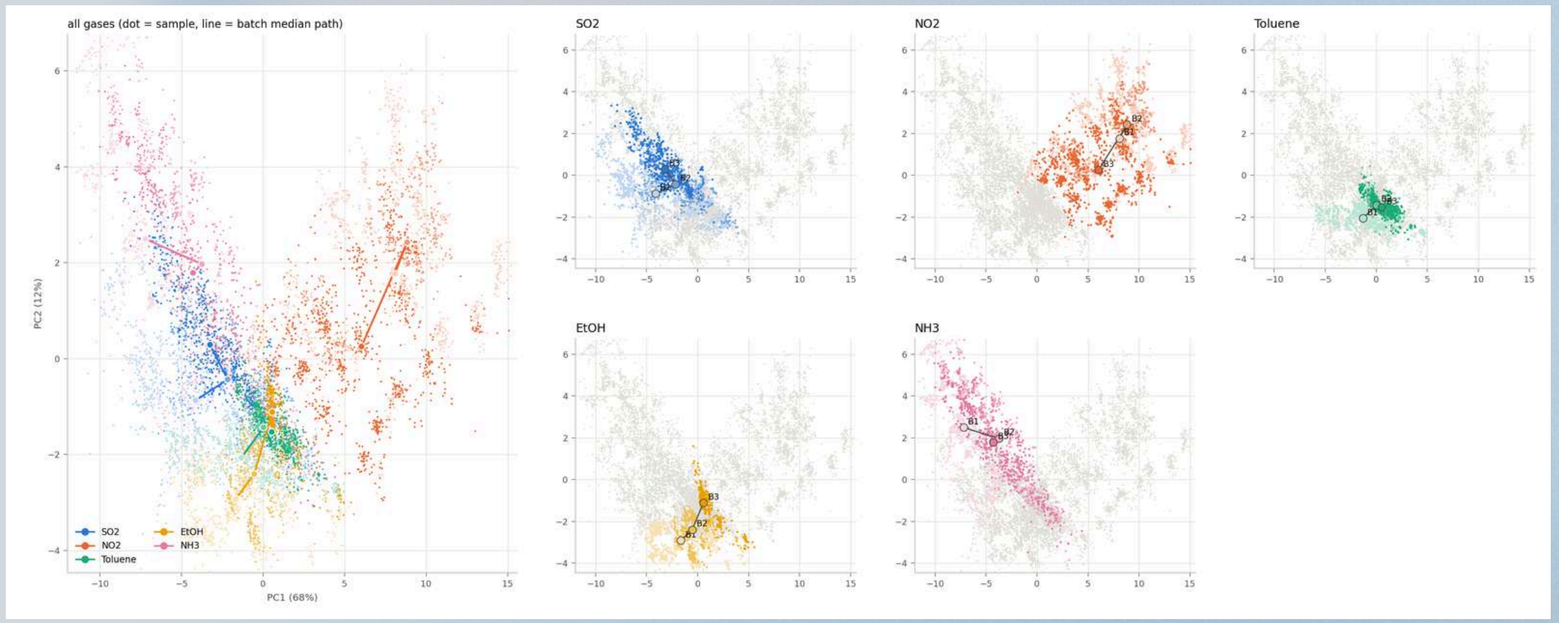
single-VOC accuracy: **design target**, not yet measured

Sensor Drift: A Persistent and Costly Problem



Why Recalibration Doesn't Scale

- Drift is not a one-time anomaly
- Calibration must be ongoing
- Every recalibration requires a complete, well-labeled dataset
- The real cost lies in how often you calibrate, not in any single calibration



MODEL · WHY A FLY BRAIN

Insect Olfaction: A Blueprint for Robust Odor Coding

Gain normalization keeps sparse coding stable

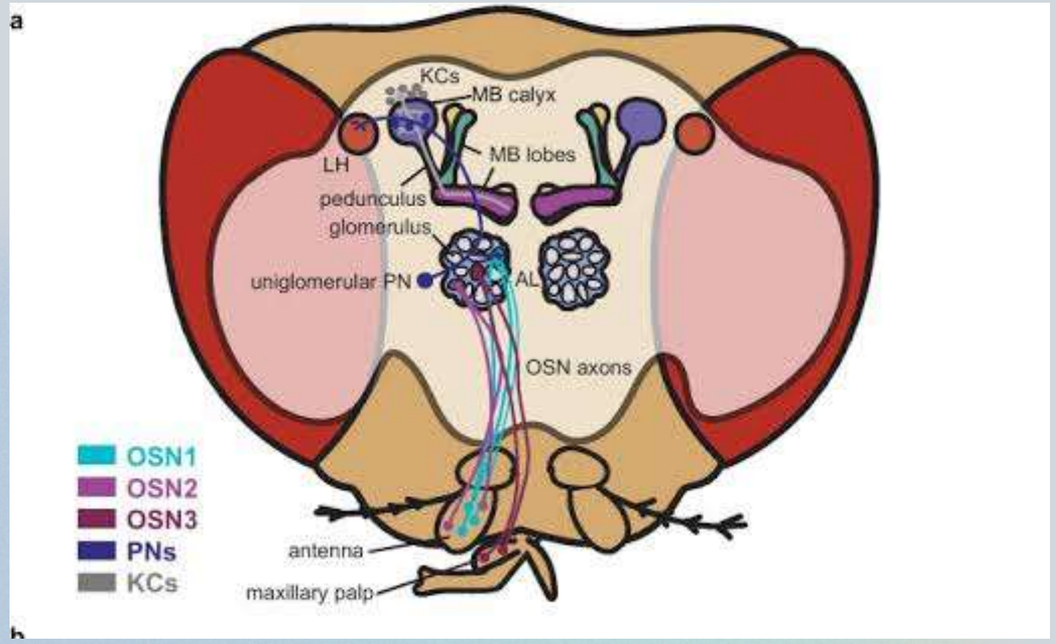
- APL normalizes Kenyon cell (KC) activity, keeping sparse odor codes stable across concentrations

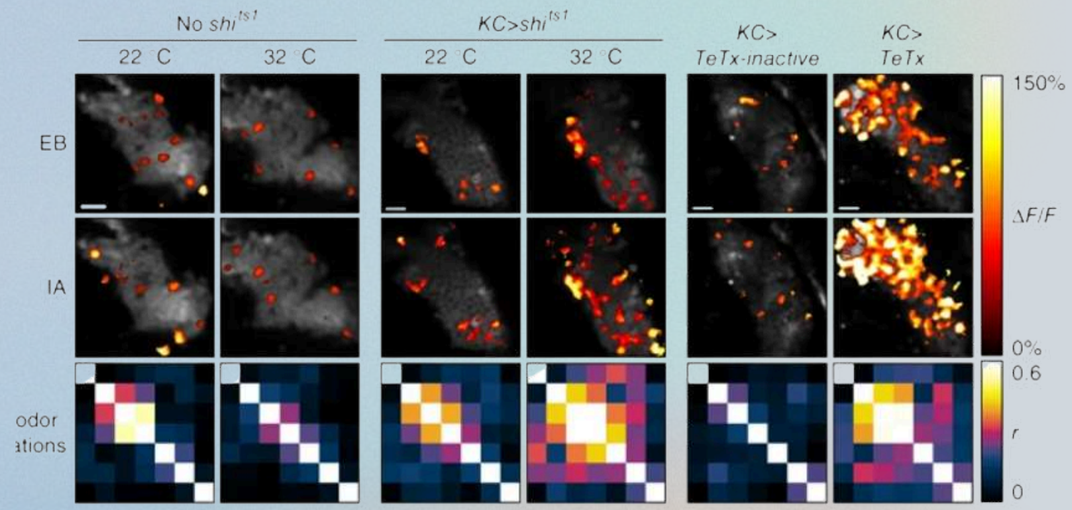
Sparse, decorrelated coding improves discrimination

- Mushroom-body sparsening and decorrelation sharpen learned odor discrimination

l inhibition and adaptation work together

level lateral inhibition plus cellular
turn dense codes into sparse,
ones





MODEL · WHY A FLY BRAIN

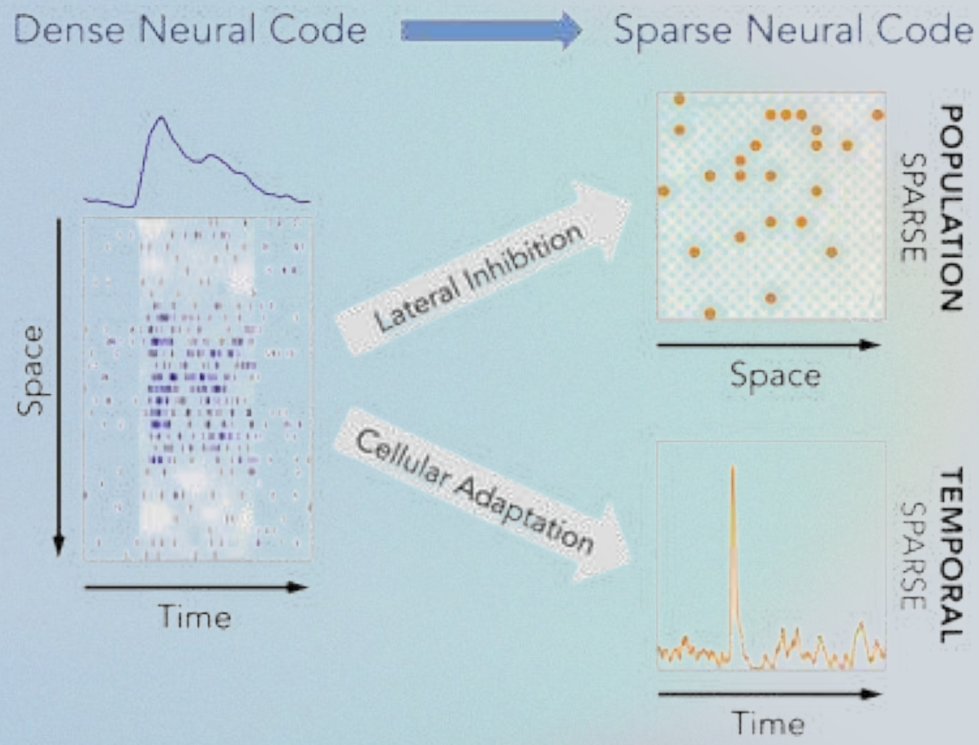
Insect Olfaction: A Blueprint for Robust Odor Coding

…tion keeps sparse coding stable

…lizes Kenyon cell (KC) activity, keeping …dor codes stable across concentrations

…se, decorrelated coding improves …crimination

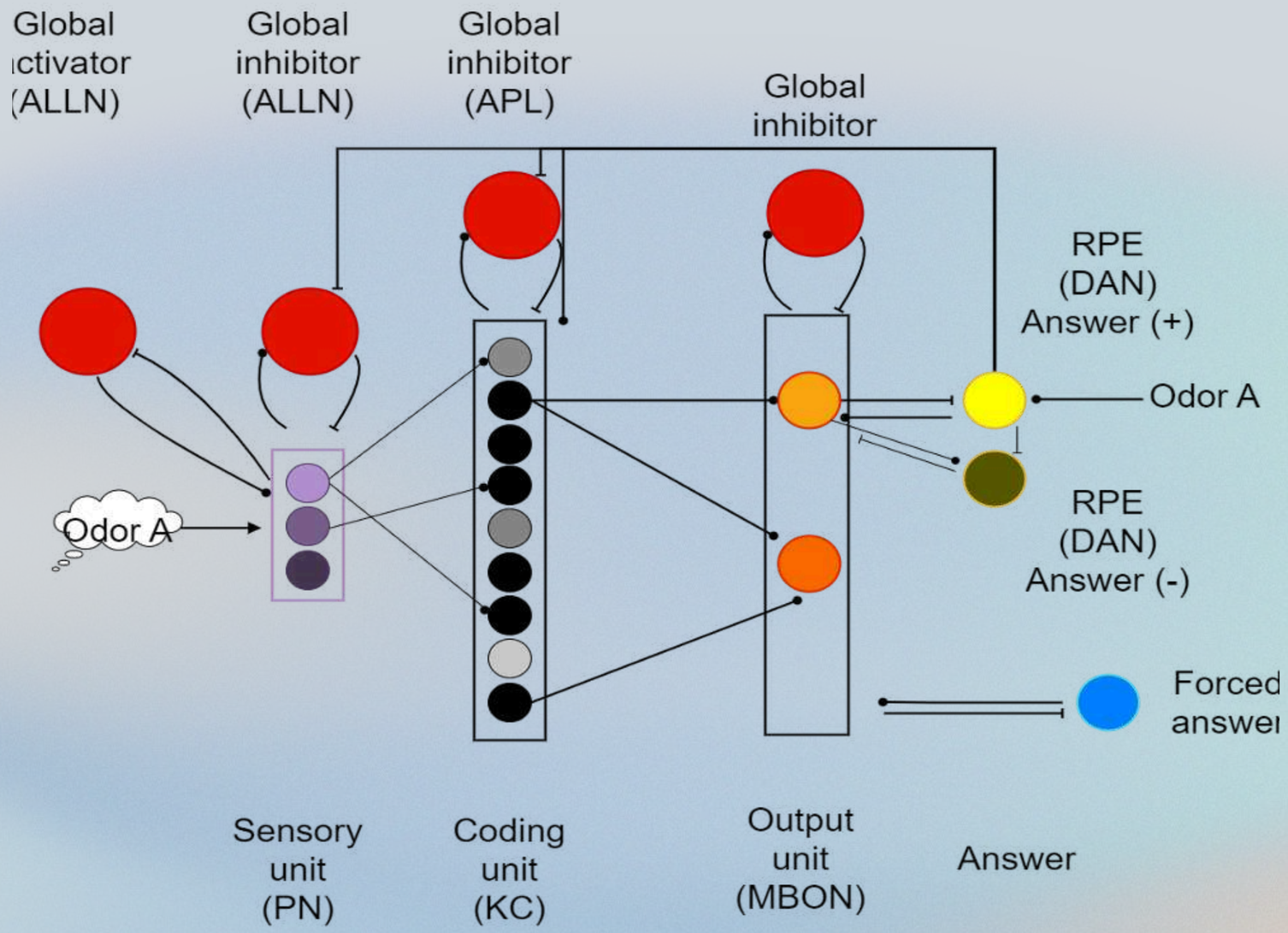
…ushroom-body sparsening and decorrelation …rpen learned odor discrimination



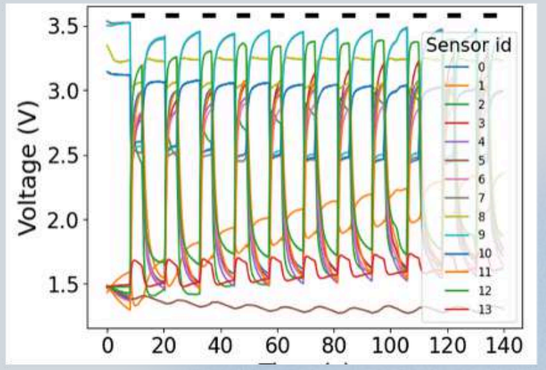
Lateral inhibition and adaptation work together

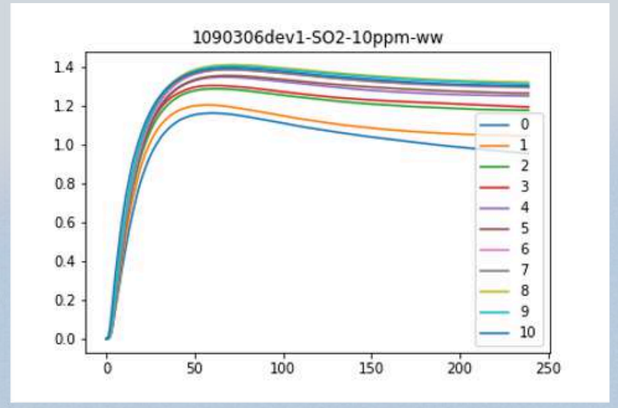
- Circuit-level lateral inhibition plus cellular adaptation turn dense codes into sparse, decorrelated ones

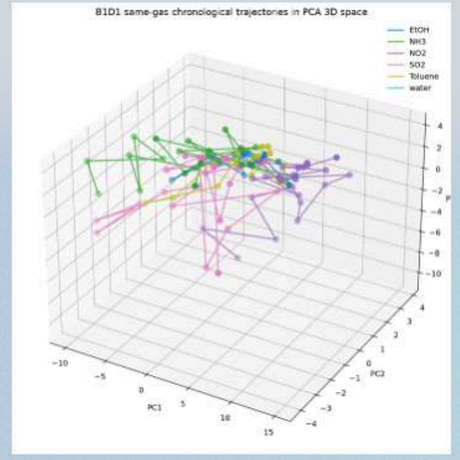
Small enough to run at the edge

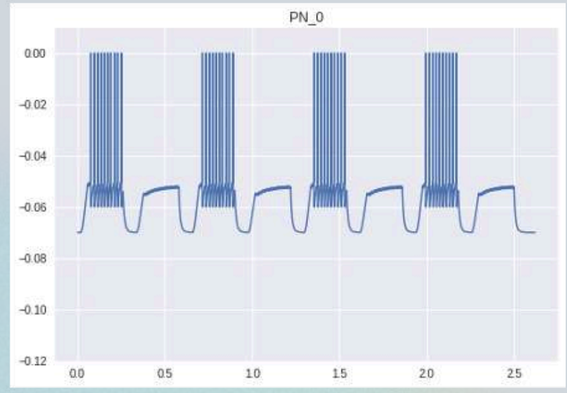


Mushroom-Body-Inspired SNN Architecture





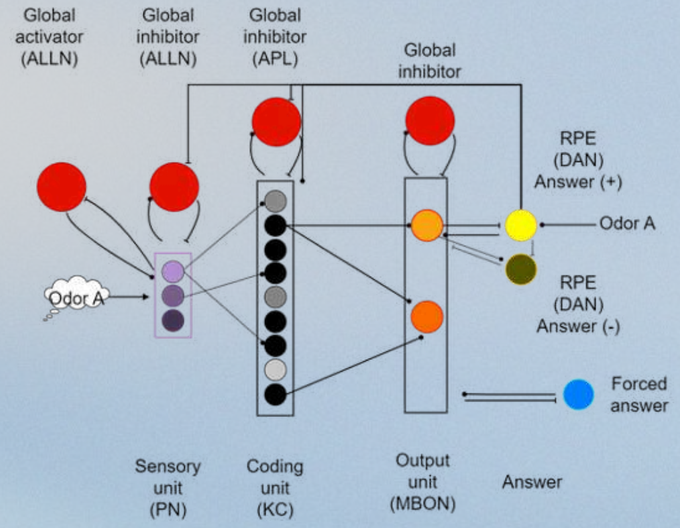




NO2, SO2, NH3, Toluene,
and EtOH; 14 electronic gas sensors

PCA for dimensionality reduction

Repeated
training

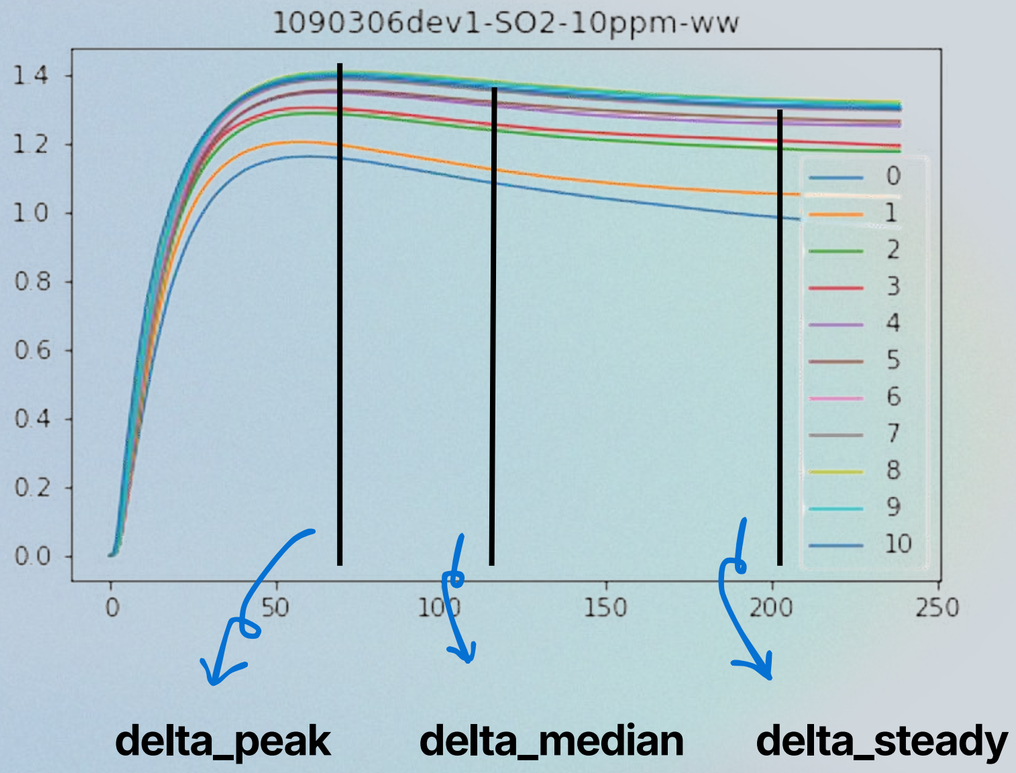


Learning mechanisms:
1. STDP
2. Synaptic homeostasis
→ weight normalization
→Pre/post
3. Intrinsic plasticity

Features of the Gas Data

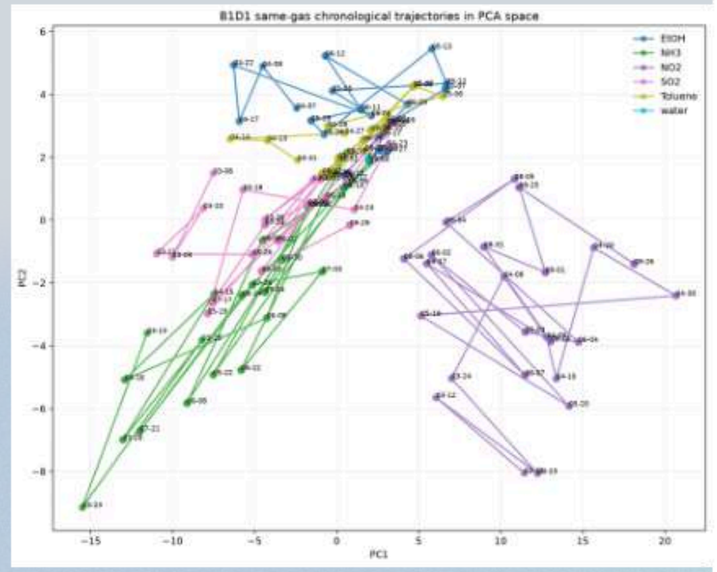
How Each Feature Is Computed

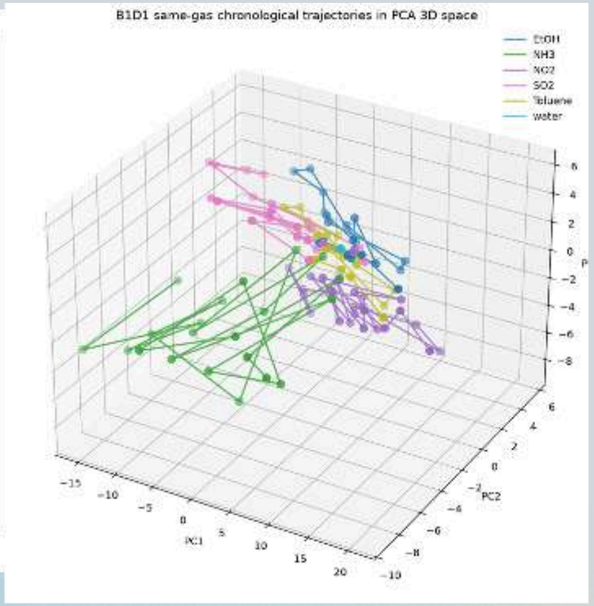
- Baseline: median of the last 60 samples of the preceding Dilute (clean air) phase, computed per sensor
- delta_median: overall shift; median over the whole Test (exposure) window minus baseline
- delta_steady: steady-state response; median of the last 60 samples of the Test window, capturing the plateau after transients settle
- delta_peak: signed peak response; largest absolute deviation from baseline during exposure, with its sign preserved
- relative_*: each of the three deltas divided by |baseline| to correct baseline offsets across sensors and devices before PCA



MODEL · WHY A FLY BRAIN

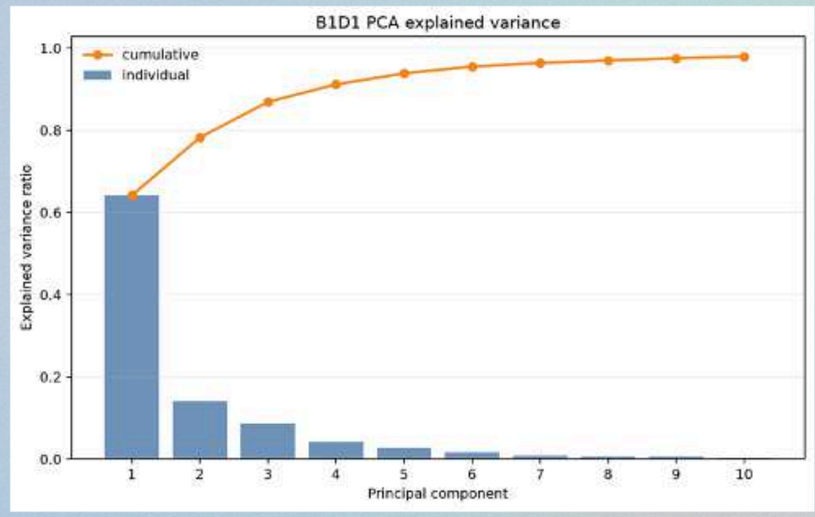
Data Preprocessing





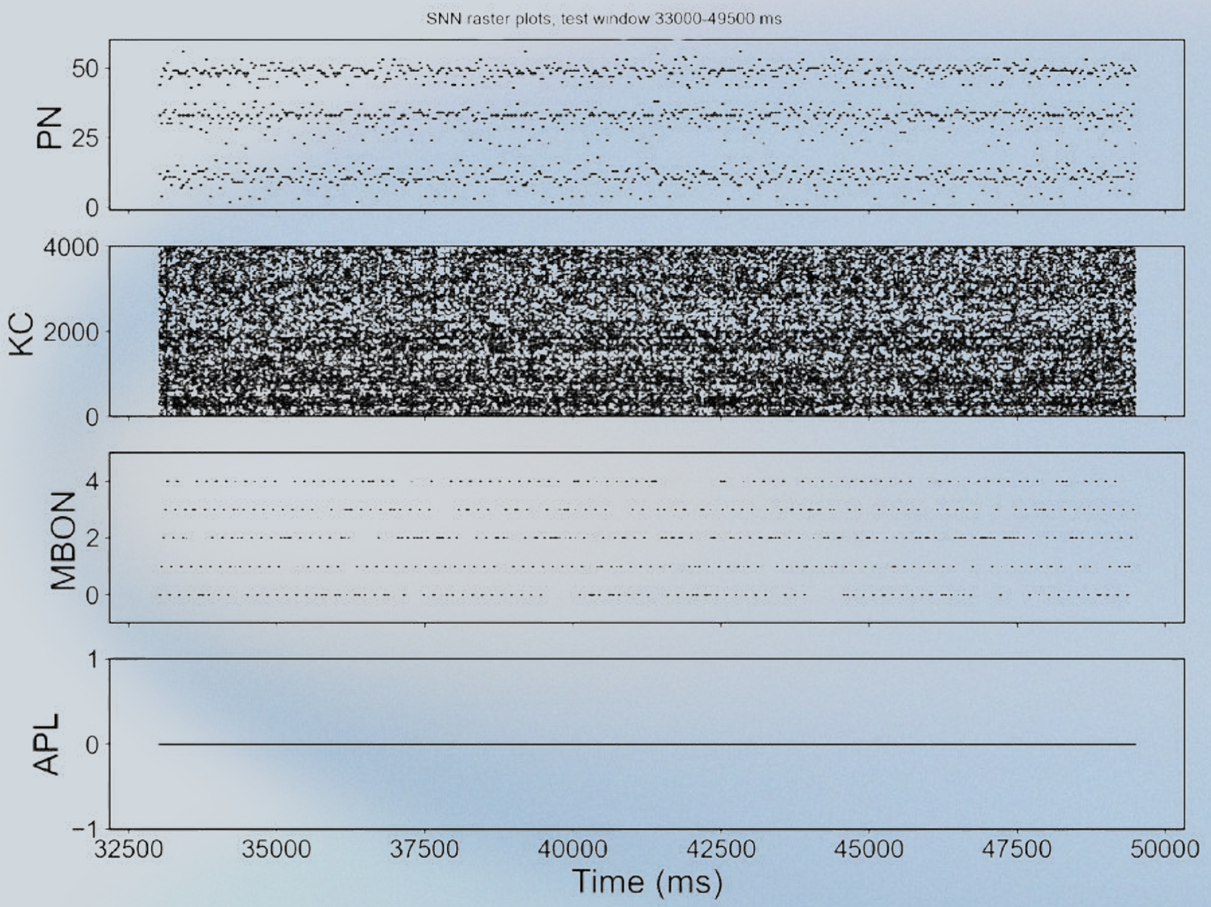
From a 14-D Sensor Space to a Fixed PN Layer

- Input: one delta_peak value per sensor, giving a 14-D vector per odor trial
- Fit on training set only: PCA is fit on training data only, then applied to both training and test sets
- 8 PCs retain ~99.5% of variance: PC1 alone explains ~74%; the first 8 PCs capture nearly all the signal (PC1=0.74, PC2=0.14, PC3=0.06; cumulative ~0.995)



MODEL · WHY A FLY BRAIN

Best SNN Performance



Parameters:

•PCA-8, p_pn_kc=0.05, w_apl_kc=30, ip=2, KC=1000, w_kc_mbon=10, lateral=2000, learning_rate=0.025

Performance:

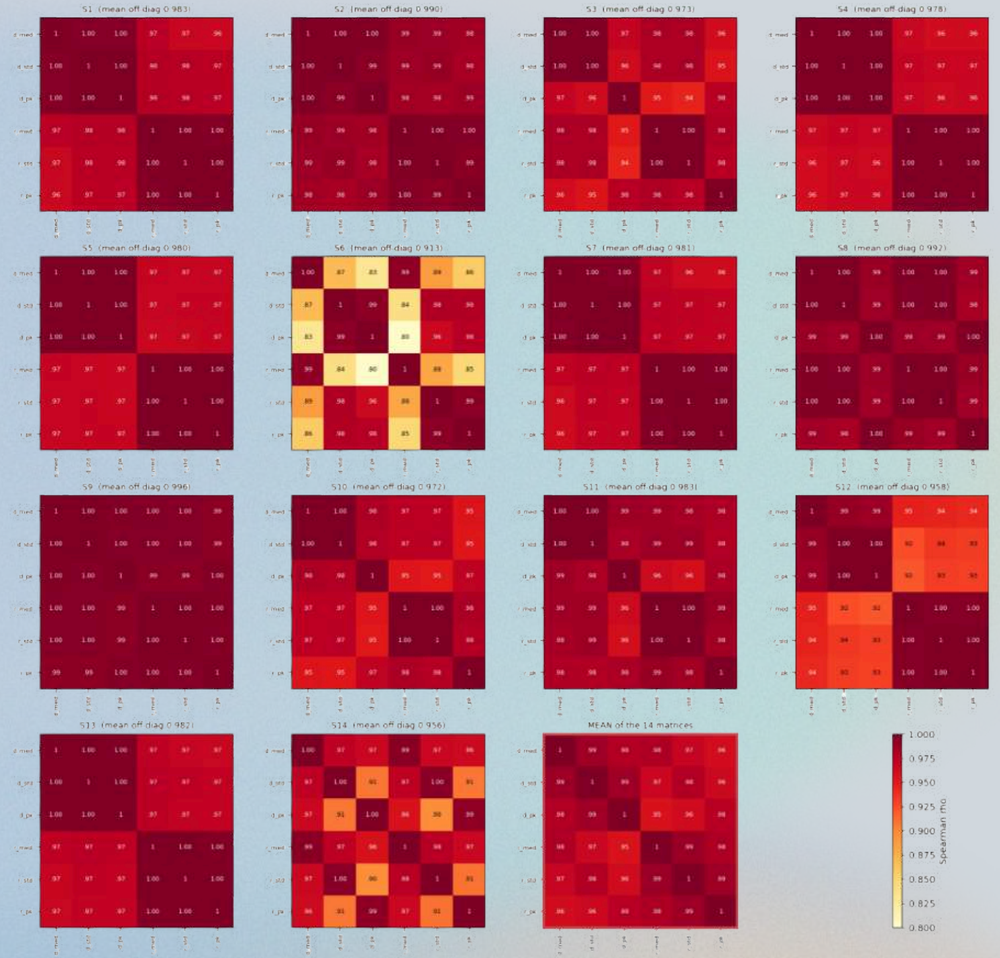
•MBON readout accuracy=0.860

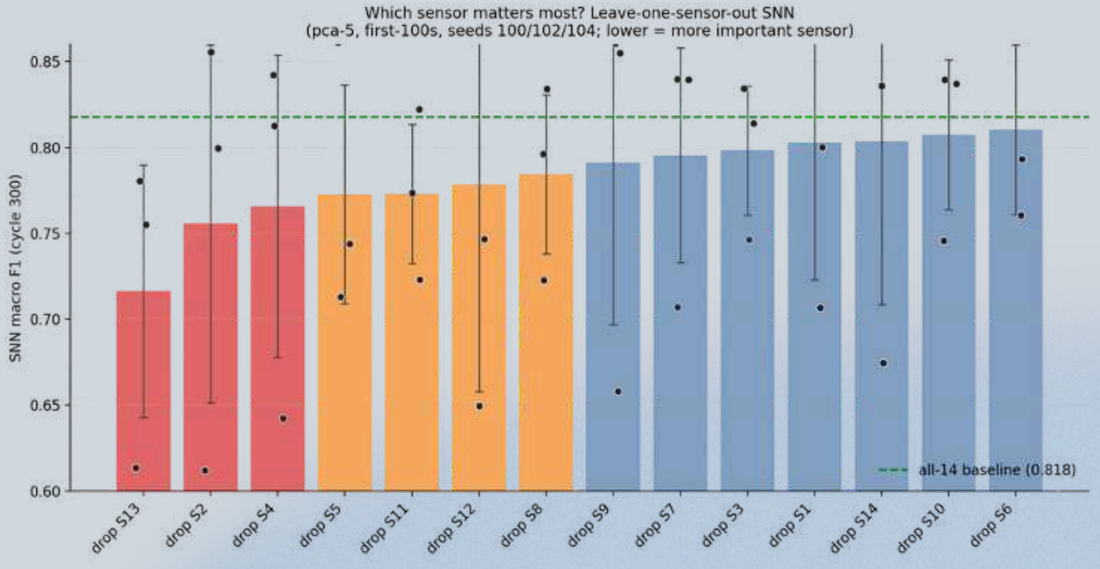
MODEL · WHY A FLY BRAIN

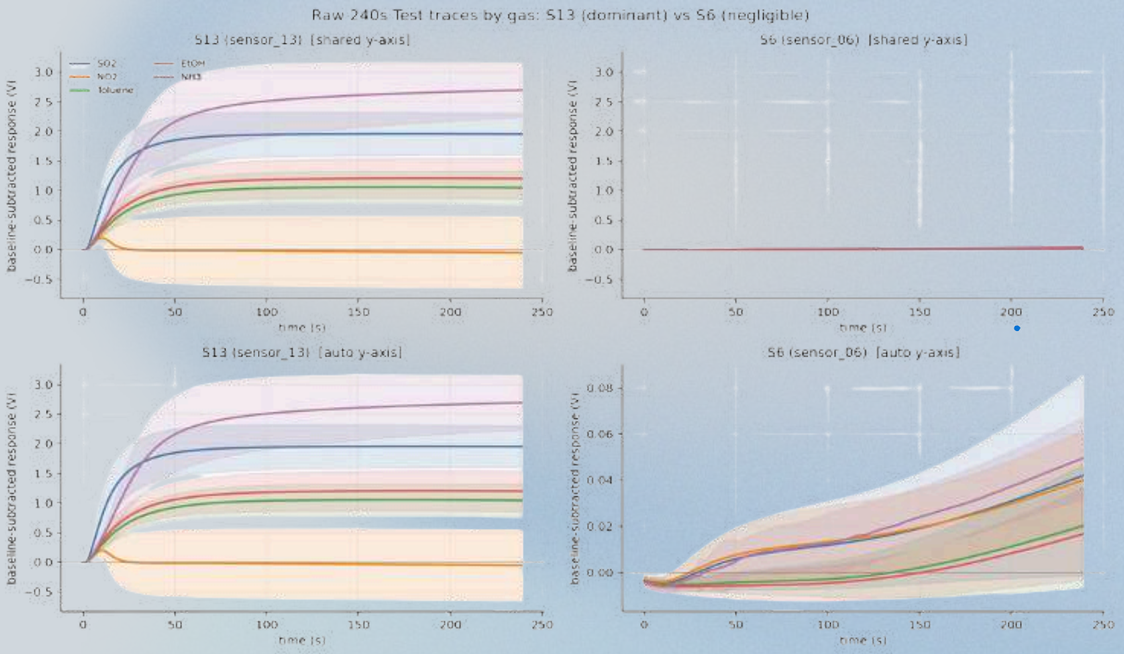
Are the 6 Features Independent?

What the Correlation Matrices Show

- Average across 14 sensors (bottom-right matrix): nearly all off-diagonal correlations exceed 0.9
- Two sensors stand out: S6 (mean ρ = 0.913) and S12 (mean ρ = 0.958) show markedly lower correlation between raw (d_*) and relative (r_*) features







MODEL · WHY A FLY BRAIN

Ablation Study

- Leave-one-sensor-out ablation: remove one sensor at a time, retrain, and compare macro F1 against the 14-sensor baseline
- S13 matters most: removing it drops F1 to ~0.72, the largest loss from any single sensor
- S6 matters least: removing it gives F1 ≈ 0.81, at or above baseline; the model loses almost nothing

Why

•The 240-s raw response curves explain the difference: S13 responds strongly and distinctly to all 5 gases, while S6's raw response is nearly flat for every gas (<0.1 V even when auto-scaled)

HARDWARE

Can we hear a faint glow in a noisy world?



HARDWARE · ENGINEERING ITERATION

Why we replaced the camera

V1 · CAMERA

Blue LED + CMOS, 30 fps

Works on the bench. Outside the lab, ambient light and sensor noise swamp a faint cell signal.

V2 · LOCK-IN

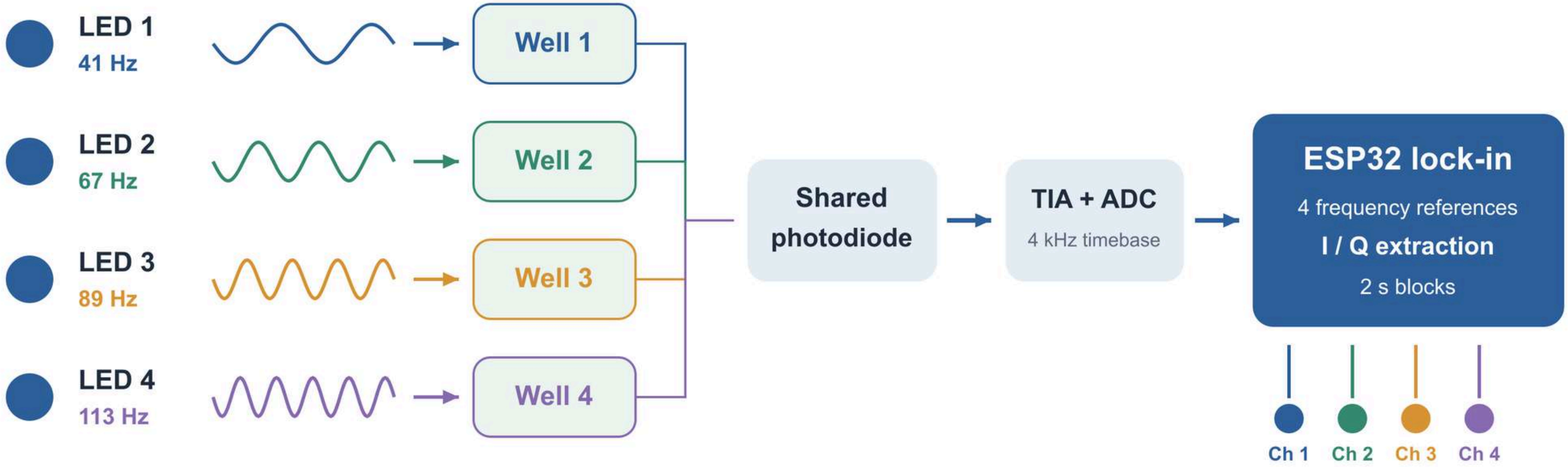
Photodiode + DLIA

Modulated LEDs with digital lock-in amplification: read only the frequency we sent, ignore the rest.

Industry told us the same thing: **sensitivity that holds in the lab often collapses in the field.**

Four excitation tags, one shared detector

The design separates frequency channels; physical well mixing still requires calibration.



Same-frequency optical leakage can survive lock-in.

Emission filtering, blanks and well calibration are still needed.

Channel colors are identifiers. All four LEDs use blue excitation; waves are illustrative, not measured.

HARDWARE · DIGITAL LOCK-IN AMPLIFICATION

Listening for one frequency

4 LEDs
each blinks at its own frequency, f1–f4

→

Cells
fluoresce in step with each LED

→

Photodiode
one detector collects all light

→

TIA + ADC
amplify and digitize

→

ESP32
lock-in recovers 4 clean channels

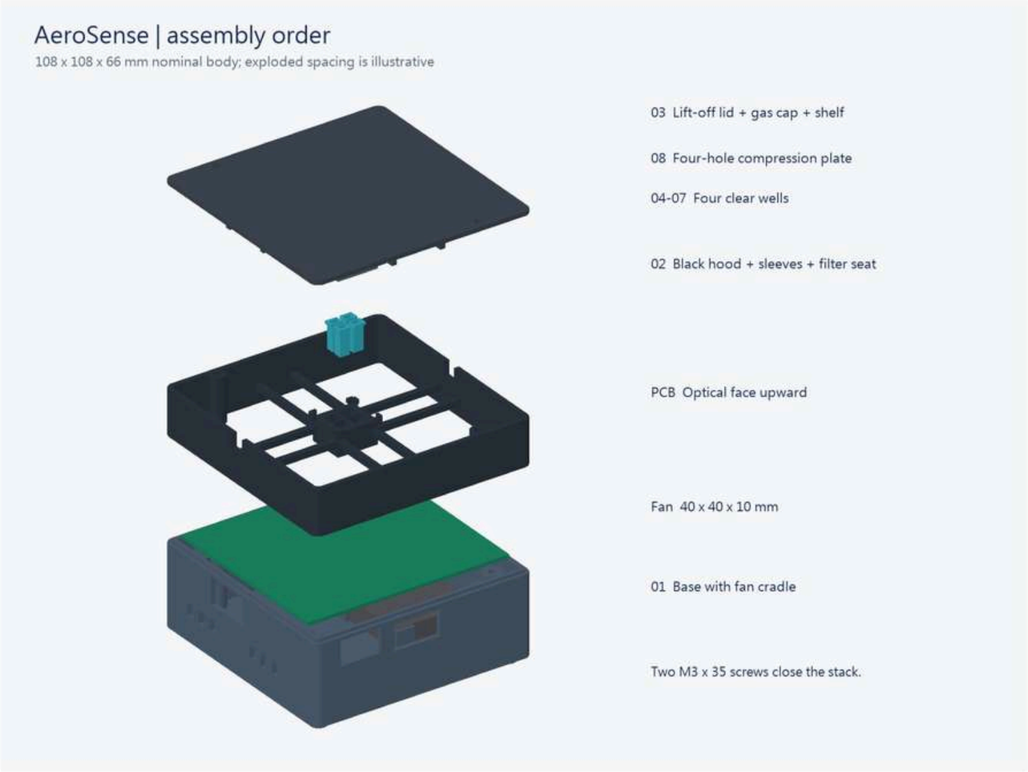
Like hearing a friend's code word at a loud party: everything off-frequency is filtered out.

Four channels at once, no moving filters, and room light and low-frequency drift are rejected.

A housing for cells, light and electronics

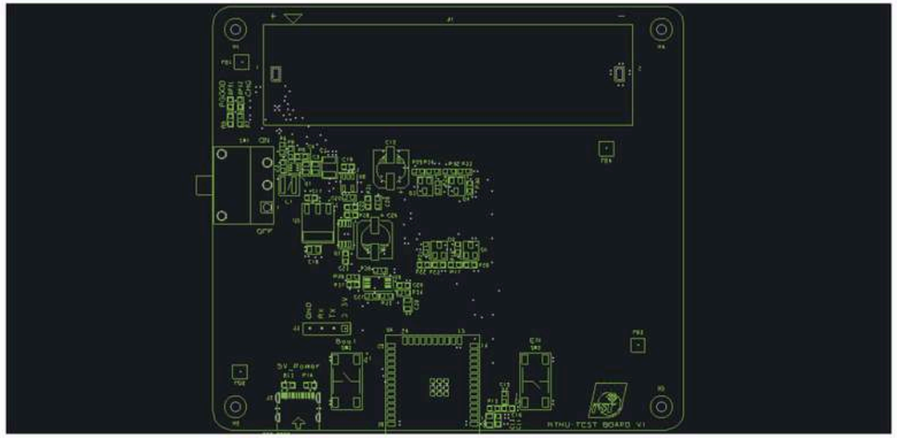
Current mechanical and electronic design sources, shown together.

Mechanical assembly



CAD drawing: exploded spacing is illustrative.

Circuit-board design



4 wells • shared gas space

108 × 108 × 66 mm nominal body

Optics → TIA → ADC → ESP32

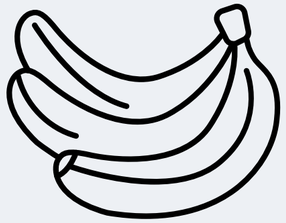
CAD and PCB layout only.
PCB fabrication and full assembly pending.

Sources: current CAD assembly_exploded.png and PCB v2 PCB_TOP.png • rendered / layout views, not device photographs

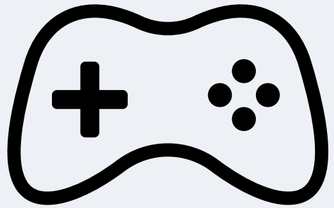
HUMAN PRACTICES

Who shaped it, and what changed?

Education



Collab with Banana Expert



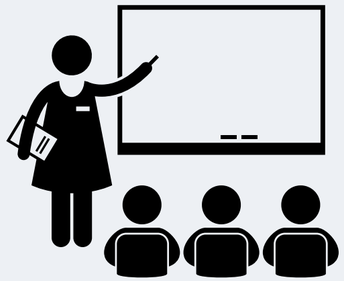
Online Games



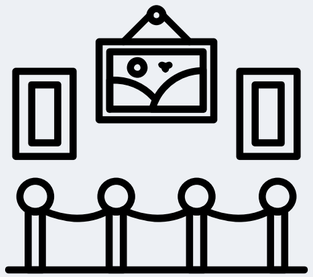
Learning Platform



Neuroscience Workshop



Elementary Camp



Art Gallery

Entrepreneurship

NTHU STARTUP GARAGE

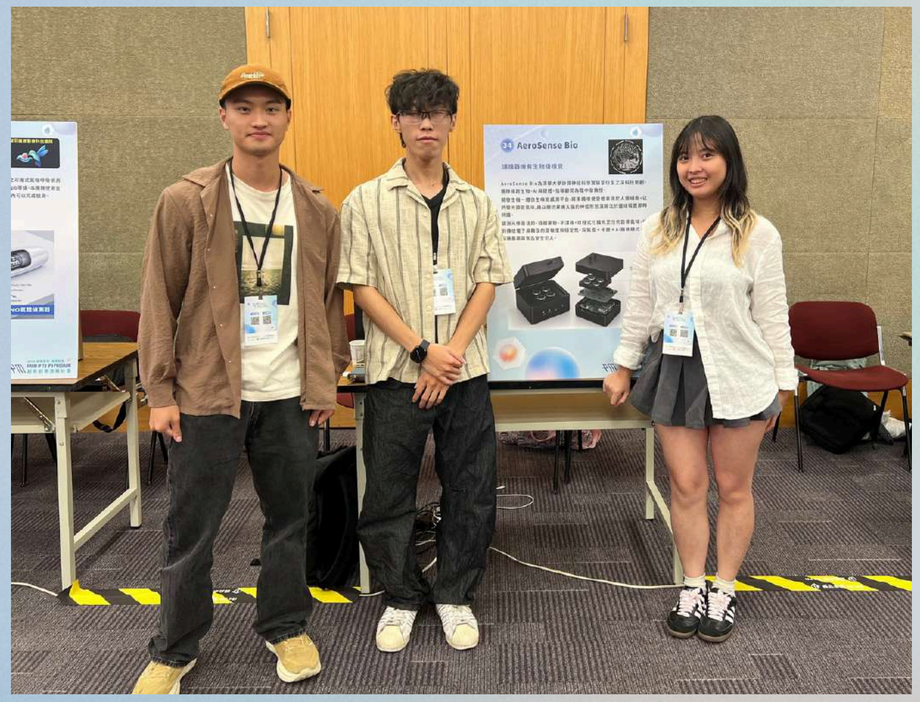




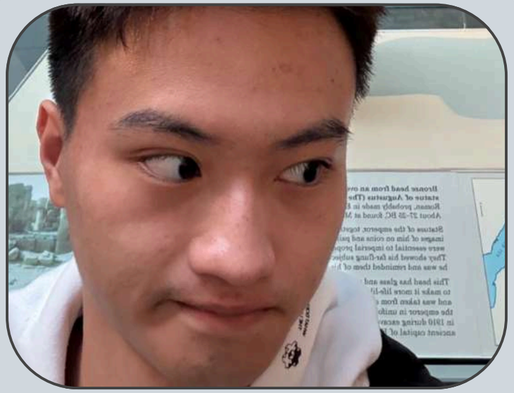
Entrepreneurship

FITI (From IP To IPO) Program by NSTC





Meet Our Team



PIN-CHE HUANG
TEAM LEADER, HW



YI-CHIA FANG
TEAM VICE LEADER, HP



CHING-CHE CHARNG
ADVISOR



CHUN-TING CHOU
MODEL



I-JUNG CHEN
WET LAB



Thank You

for your time
and attention

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